

8 DEALING WITH DATABASES

8.1 Understanding the WordPress database class

If you have developed Web sites or applications with PHP for any length of time, you are probably familiar with PHP database functions such as `mysql_query()`, `mysql_fetch_array()`, and so on. You might also be familiar with database abstraction classes. Like other abstraction classes, the database class used by WordPress provides an abstraction layer between the developer and the raw MySQL functions in PHP.

A database abstraction layer is a class or library that sits between your code and a series of databases. As every database has its own nuances, and the built-in database functions are messy and inconsistent, a database abstraction layer sorts out that messiness by providing a consistent interface to multiple databases.

The database class used by WordPress — `ezSQL` — is an extremely lightweight MySQL database abstraction class that, like much of WordPress' core, originated elsewhere. Before diving too far in, you should take note that `ezSQL` is not a database abstraction layer; it is a MySQL abstraction layer. It is possible, however, to take the `wp-db.php` file and manipulate it to talk to other database systems such as PostgreSQL or Microsoft's SQL Server. If you do this, simply save the file as `db.php` and drop it in the `wp-includes/` folder. WordPress looks to see if a `db.php` file exists first when bootstrapping, the process of getting WordPress connected to the database, and will use it instead of `wp-db.php` if it exists. You still need to know and understand MySQL or your underlying database. The class merely makes it easier to leverage.

`ezSQL` is not a full-featured abstraction engine like, for example, PHP 5's PHP Data Objects (PDO) library. It is a lightweight class that proves useful in many aspects of PHP development. It is perfectly suited to use within WordPress based on its licensing, size, and speed.

8.2 Examining the Schema and Properties of the database class

Because the database class is an object, it has quite a few properties and methods. Understanding these is critical for plugin developments because you will be using the database a lot. The database object is assigned to the handler `$wpdb` throughout WordPress. Whenever you create a function that requires transactions with the database, excluding the use of core